

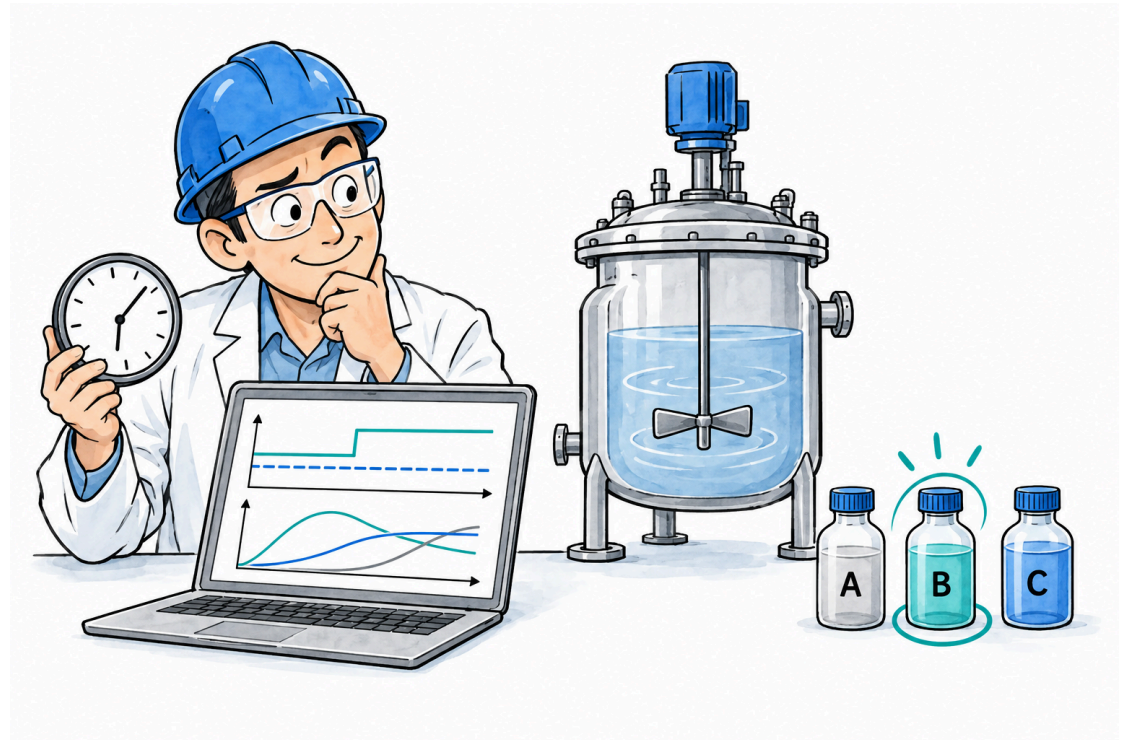
Week 2: Learning Multicomponent Steady-State Responses with MLPs

More conversion does not tell the whole composition

The desired intermediate is **B** in $A \rightarrow B \rightarrow C$. At short residence time little B forms; at long residence time more B reacts onward to C.

An engineer needs all three outlet concentrations at each condition. One conversion number does not describe the product composition.

Can one model predict A, B, and C consistently?



Three outputs, different errors

All three concentrations have the same units, but their variation differs. An unscaled loss can emphasize one component.

This week learns how an MLP computes these outputs, how one gradient update changes its parameters, and how to evaluate each component.

Learn and check the steady-state MLP

1. Derive steady-state labels for A, B, and C.
2. Scale the data and follow a forward pass, backpropagation, and one parameter update.
3. Train the MLP and compare component RMSE and concentration-sum residuals.

1. $A \rightarrow B \rightarrow C$ in a CSTR

Consider a perfectly mixed, constant-volume, constant-density, isothermal CSTR with consecutive first-order reactions $A \rightarrow B \rightarrow C$. The feed contains A; its B and C concentrations are zero. Temperature is an operating input. With concentrations in mol/L, temperature in K, and time in min, the balances are:

$$\begin{aligned}\frac{dC_A}{dt} &= \frac{C_{Af} - C_A}{\tau} - k_1 C_A \\ \frac{dC_B}{dt} &= -\frac{C_B}{\tau} + k_1 C_A - k_2 C_B \\ \frac{dC_C}{dt} &= -\frac{C_C}{\tau} + k_2 C_B\end{aligned}$$

Rate constants and operating domain

Both rate constants have units of 1/min. Use the following temperature dependence:

$$k_1(T) = 0.2 \exp\left[6000\left(\frac{1}{330} - \frac{1}{T}\right)\right]$$

$$k_2(T) = 0.1 \exp\left[5000\left(\frac{1}{330} - \frac{1}{T}\right)\right]$$

Quantity	Value or range
Reference temperature	330 K
Reference k_1 ; k_2	0.2; 0.1 per minute
Temperature-sensitivity parameters	6000 K; 5000 K
Temperature T	300–360 K
Residence time τ	1–10 min
Feed A concentration	0.8–1.5 mol/L

The three balances describe A consumption, B formation and consumption, and C formation. Adding them cancels the reaction terms.

2. Derive the three-output steady-state map

Set the time derivatives to zero and multiply each balance by τ . Collect the concentration terms:

$$\begin{aligned}(1 + k_1\tau)C_A &= C_{Af} \\ (1 + k_2\tau)C_B &= k_1\tau C_A \\ C_C &= k_2\tau C_B\end{aligned}$$

Solve for A, B, and C

Solve in the order A, B, C:

$$\begin{aligned}C_A &= \frac{C_{Af}}{1 + k_1\tau} \\C_B &= \frac{k_1\tau C_{Af}}{(1 + k_1\tau)(1 + k_2\tau)} \\C_C &= k_2\tau C_B\end{aligned}$$

At 330 K, $\tau = 5$ min, and feed = 1.2 mol/L, the concentrations are **[0.6, 0.4, 0.2] mol/L**. Their sum equals the feed concentration. These three outputs are the labels for the steady-state MLP.

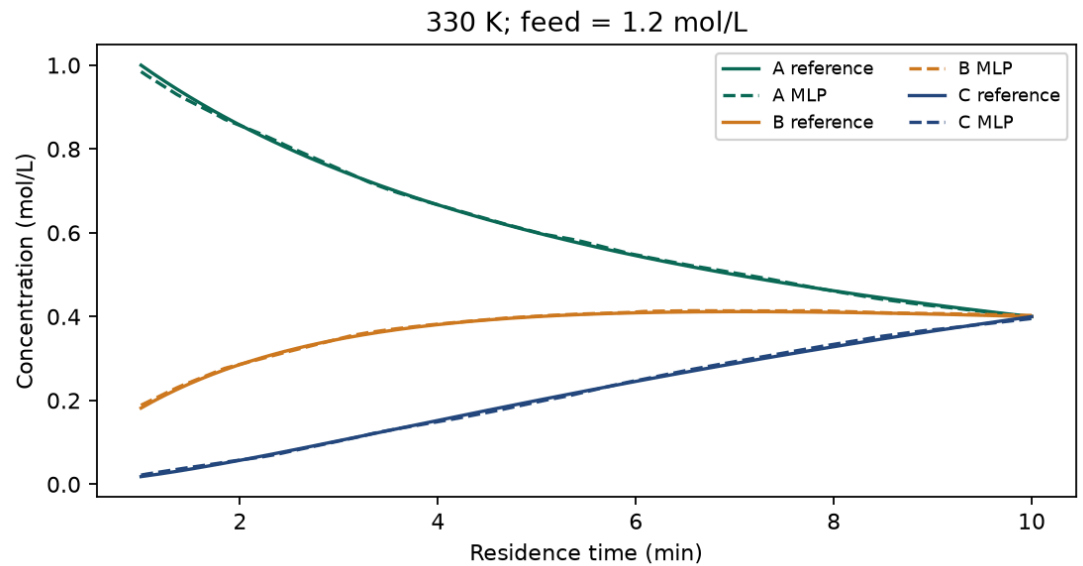
Why the B response is nonmonotone

At a fixed temperature, the B yield and its maximizing residence time follow from the expression for C_B:

$$Y_B = \frac{C_B}{C_{Af}} = \frac{k_1 \tau}{(1 + k_1 \tau)(1 + k_2 \tau)}, \quad \tau^* = \frac{1}{\sqrt{k_1 k_2}}$$

Compare the three concentration profiles

At 330 K, τ^* is about 7.07 min. Short residence times limit B formation; long residence times allow more B to react to C.



Three-component response at 330 K. The B concentration has an interior maximum.

3. Prepare the supervised dataset

Each row contains inputs $[T, \tau, C_{Af}]$ and outputs $[C_A, C_B, C_C]$. Input and output arrays both have shape $(N, 3)$. Generate 400 train, 120 validation, and 160 test operating points within the table's bounds, then calculate their labels with the steady-state map.

The three concentrations have the same units but different variations. In the training data, their standard deviations are approximately **[0.283, 0.110, 0.217] mol/L**.

4. Forward calculation and scaling

Use a $3 \rightarrow 32 \rightarrow 32 \rightarrow 3$ MLP, with ReLU hidden layers and a linear output layer:

Each hidden layer first computes a weighted sum and bias. $\text{ReLU}(a) = \max(0, a)$ retains positive values and sets negative values to zero. The weights and biases are the parameters fitted to the concentration data.

$$h_1 = \text{ReLU}(W_1 \tilde{x} + b_1)$$

$$h_2 = \text{ReLU}(W_2 h_1 + b_2)$$

$$\hat{z} = W_3 h_2 + b_3$$

Layer	Weight shape	Bias shape	Parameters
Input $\rightarrow$ hidden 1	32×3	32	128
Hidden 1 $\rightarrow$ hidden 2	32×32	32	1056
Hidden 2 $\rightarrow$ output	3×32	3	99
Total			1283

Scaling and the training loss

Scale the inputs with the declared lower and upper bounds. Standardize each output with its training mean μ and standard deviation s , and restore physical units after prediction:

$$\tilde{x}_j = 2 \frac{x_j - l_j}{u_j - l_j} - 1$$
$$z_j = \frac{y_j - \mu_j}{s_j}, \quad \hat{y}_j = \mu_j + s_j \hat{z}_j$$

The loss averages squared errors across samples and standardized output components:

$$L = \frac{1}{3N} \sum_{n=1}^N \sum_{j=1}^3 \left(\hat{z}_{n,j} - z_{n,j} \right)^2$$

Thus a concentration error is measured relative to that component's training variation. For example, the smaller B standard deviation gives its raw squared error a larger weight in this loss.

5. Backpropagation: one calculation

Take a $3 \rightarrow 2 \rightarrow 3$ network with zero biases and the following values. For this calculation, use $L = \frac{1}{2}\|\hat{y} - y\|^2$ and learning rate 0.1.

$$x = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, y = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$
$$W_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, W_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

The hidden pre-activation a , hidden activation h , prediction, and error are:

$$a = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, h = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \hat{y} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, e = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Gradients and the parameter update

The output error propagates through the output weights and the ReLU derivative:

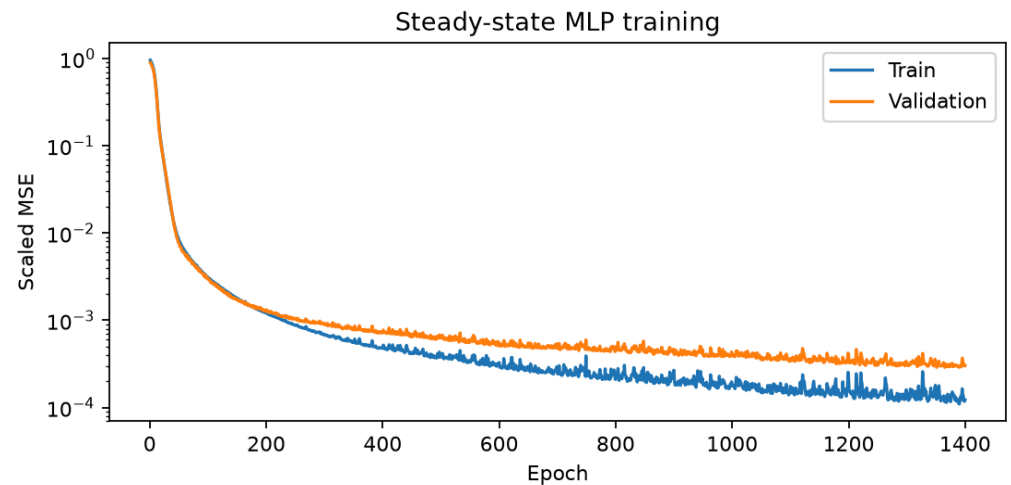
$$\delta_2 = e, \nabla_{W_2} L = e h^T = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 1 & 0 \end{bmatrix}$$
$$\delta_1 = \left(W_2^T e \right) \odot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$
$$\nabla_{W_1} L = \delta_1 x^T = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

The bias gradients are δ_1 and δ_2 . Update every weight and bias with $\text{parameter} \leftarrow \text{parameter} - 0.1 \times \text{gradient}$. The next forward calculation gives **[0.26, 0.14, 0.26]**, and the loss decreases from **1.5** to **0.4374**. The notebook checks these gradients using autograd.

6. Train and evaluate the steady-state model

Use Adam with learning rate 0.001, batches of 64, and up to 1400 epochs. After each epoch, evaluate validation loss and retain the weights at its minimum. One batch follows these operations:

```
prediction = model(inputs)
loss = loss_function(prediction, targets)
optimizer.zero_grad()
loss.backward()
optimizer.step()
```

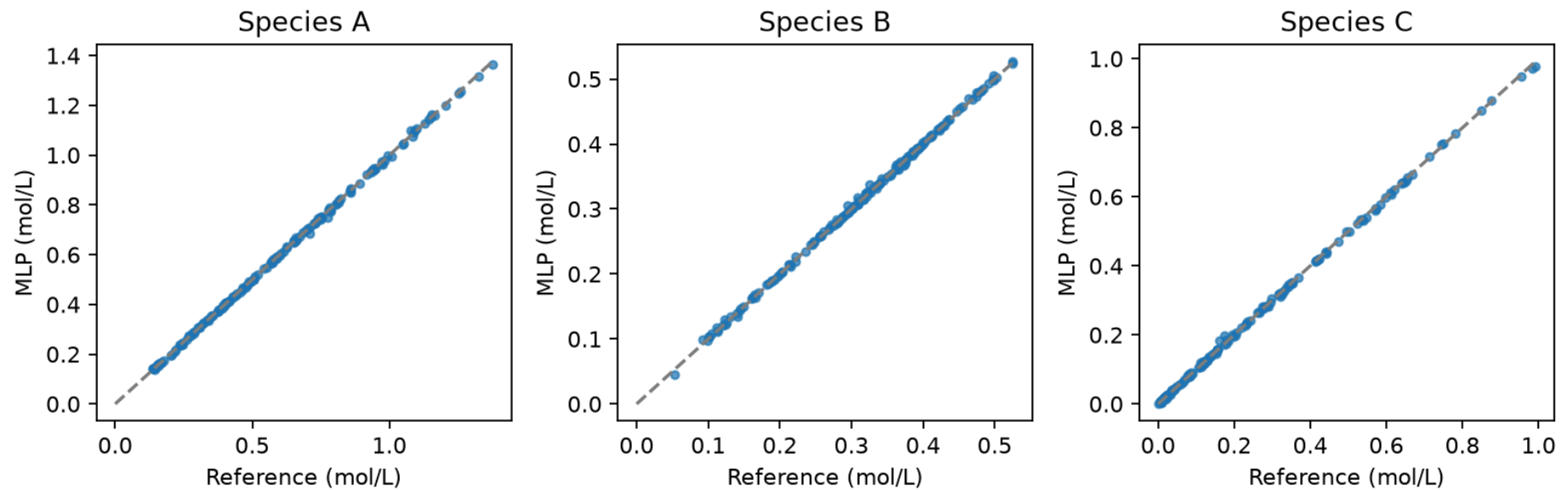


Training and validation loss. Prediction metrics below are computed after restoring mol/L units.

Evaluate each output component

Test RMSE, mol/L	A	B	C
Output scaling	0.005275	0.002940	0.004423
Unscaled outputs	0.006634	0.003917	0.004162

In this comparison, output scaling reduces A and B RMSE; C RMSE is slightly higher. Evaluate each component rather than combining all three into one error number.



Separate parity plots reveal each component's errors.

Check the concentration sum

Also calculate the concentration-sum residual:

$$r = \hat{C}_A + \hat{C}_B + \hat{C}_C - C_{Af}$$

Its test RMSE is **0.003781 mol/L**. Component prediction errors and the residual describe different aspects of the model.

The three linear outputs are free predictions. Balance-consistent training labels do not force their sum to match the feed concentration.

For feed **1.2 mol/L**, predictions **[0.62, 0.41, 0.20] mol/L** sum to **1.23 mol/L**, giving **r = 0.03 mol/L**. A model predicting mole fractions can likewise return a sum different from one.

A small prediction error does not enforce a required equality.

Week 4 preview: put the constraint into the model

We can correct a prediction after training, or include a constraint-respecting transformation in the training calculation.

Approach	Where the correction enters
Post-training mapping/projection	Train the raw predictor, then correct its outputs.
KKT-hPINN	Apply a fixed affine projection using the input and raw prediction before the loss, and at inference.

$$x \rightarrow f_{\theta} \rightarrow \hat{y} \rightarrow \Pi_x \rightarrow \tilde{y} \rightarrow \text{loss}$$

KKT-hPINN computes loss on the projected output and backpropagates through the projection. Its coefficients are fixed; the MLP weights learn with that transformation in place.

Week 4 derives the projection and KKT conditions, checks rank assumptions, and follows the gradient. The guarantee concerns the specified linear equalities, up to numerical arithmetic; nonnegativity needs an additional constraint.

Source: Chen et al. (2024), [DOI: 10.1016/j.compchemeng.2024.108764](https://doi.org/10.1016/j.compchemeng.2024.108764). [Open-access version: Section 3 and Remark 3](#).

7. Run the required steady-state lab

Download the [steady-state notebook](#) or [lab bundle](#). The bundle also preserves optional later-topic materials.

```
python -m pip install -r requirements.txt  
python week02_lab.py --mode steady --output-dir week02-results
```

Evaluate the trained predictor by component RMSE and concentration-sum residual. The stored operating-selection results belong to the later optimization part.

8. Core exercises and submission

1. Derive C_A , C_B , and C_C and verify their sum. Explain the nonmonotone B curve.
2. Count the **1283** steady-state MLP parameters and verify the hand-calculated gradient update.
3. Compare scaled and unscaled outputs by each component's RMSE.
4. Inspect concentration-sum residual and minimum predicted concentration on held-out inputs.

Submit the steady-state model, split/scaling details, checked gradient update, and component/consistency evaluation.

Original explanations: PyTorch [model construction](#), [autograd](#), and [training](#).

Reading companion

The [Week 2 companion](#) and [English PDF](#) explain the three-output map, parameter count, scaling, gradient update, training loop, and independent diagnostics.

KKT-hPINN reference: Chen, H., Constante Flores, G. E., & Li, C. (2024). [Physics-informed neural networks with hard linear equality constraints](#). *Computers & Chemical Engineering*, **189**, 108764.

Generate dynamic concentration data

Use the same balances to generate 30 min trajectories at $\Delta t = 0.2$ min. There are 150 input intervals and 151 concentration times. Feed concentration and residence time are fixed within each trajectory; temperature can change between intervals. Each initial concentration vector sums to the feed concentration.

Adding the balances gives the evolution of total concentration:

$$\frac{d(C_A + C_B + C_C)}{dt} = \frac{C_{Af} - (C_A + C_B + C_C)}{\tau}$$

With this initial condition and fixed feed concentration, the reference concentration sum stays equal to C_{Af} throughout the trajectory.

Generate **60 train, 15 validation, and 20 test trajectories**. Their transition counts are 9000, 2250, and 3000. Assign whole trajectories to the three sets. Each transition row contains the current concentrations, operating inputs, and the next concentrations.

Learn a dynamic MLP

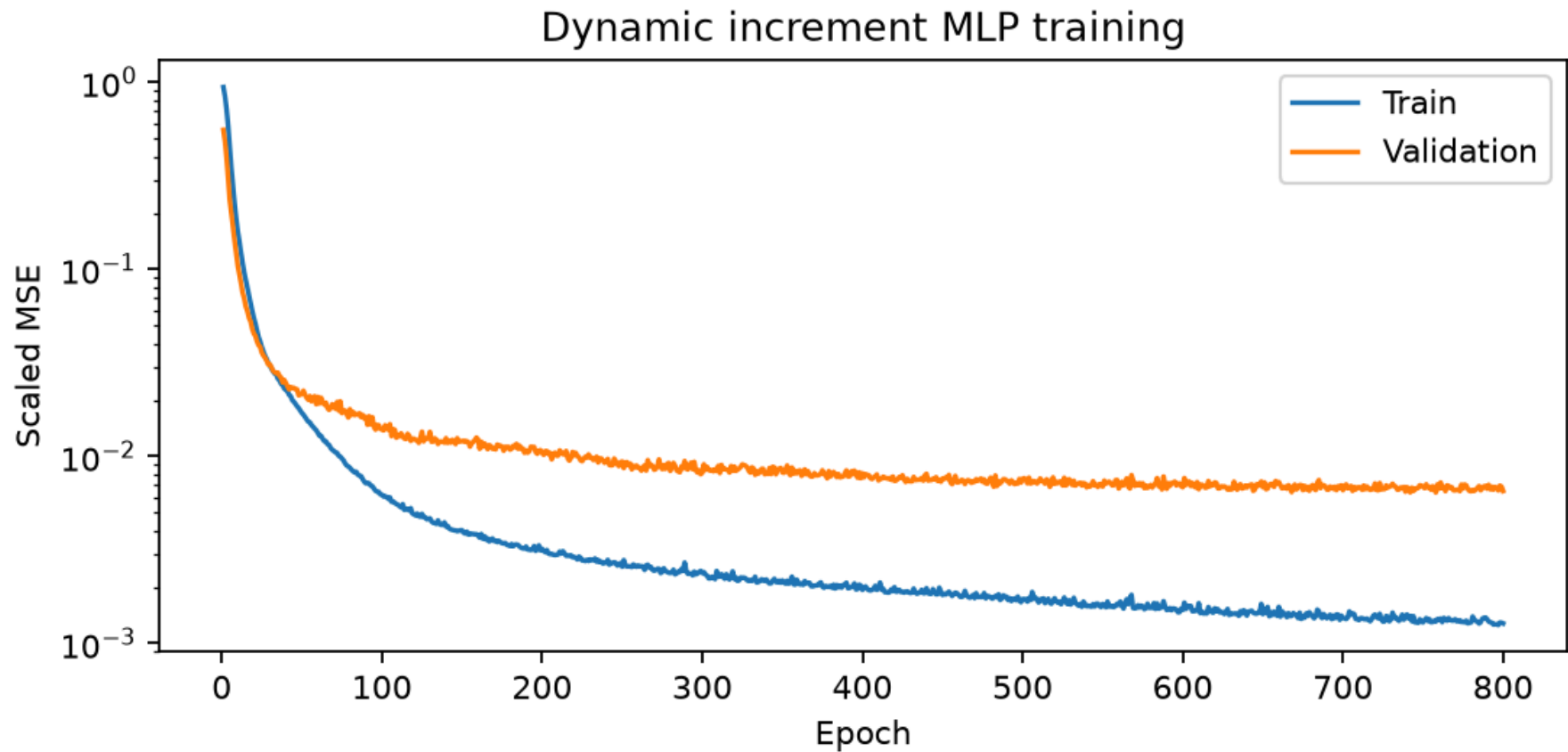
The network receives six values: three current concentrations and $[T, \tau, C_{Af}]$. Use a $6 \rightarrow 32 \rightarrow 32 \rightarrow 3$ architecture, containing **1379 parameters**. Train its three outputs on concentration increments $\Delta c = c_{\text{next}} - c_{\text{current}}$. Standardize both inputs and increments with training statistics.

$$v_k = \begin{bmatrix} C_{A_k} \\ C_{B_k} \\ C_{C_k} \\ T_k \\ \tau_k \\ C_{Af} \end{bmatrix}$$

$$\hat{c}_{k+1} = c_k + \mu_{\Delta} + s_{\Delta} \odot g_{\theta}(\tilde{v}_k)$$

Here μ_{Δ} and s_{Δ} restore the increment to mol/L units. Adding it to the current concentration produces the next state. Train with Adam, learning rate 0.001, batches of 512, and up to 800 epochs; retain the validation-minimum weights.

Dynamic model training history



The dynamic loss is measured on standardized increments.

One-step prediction and time-series rollout

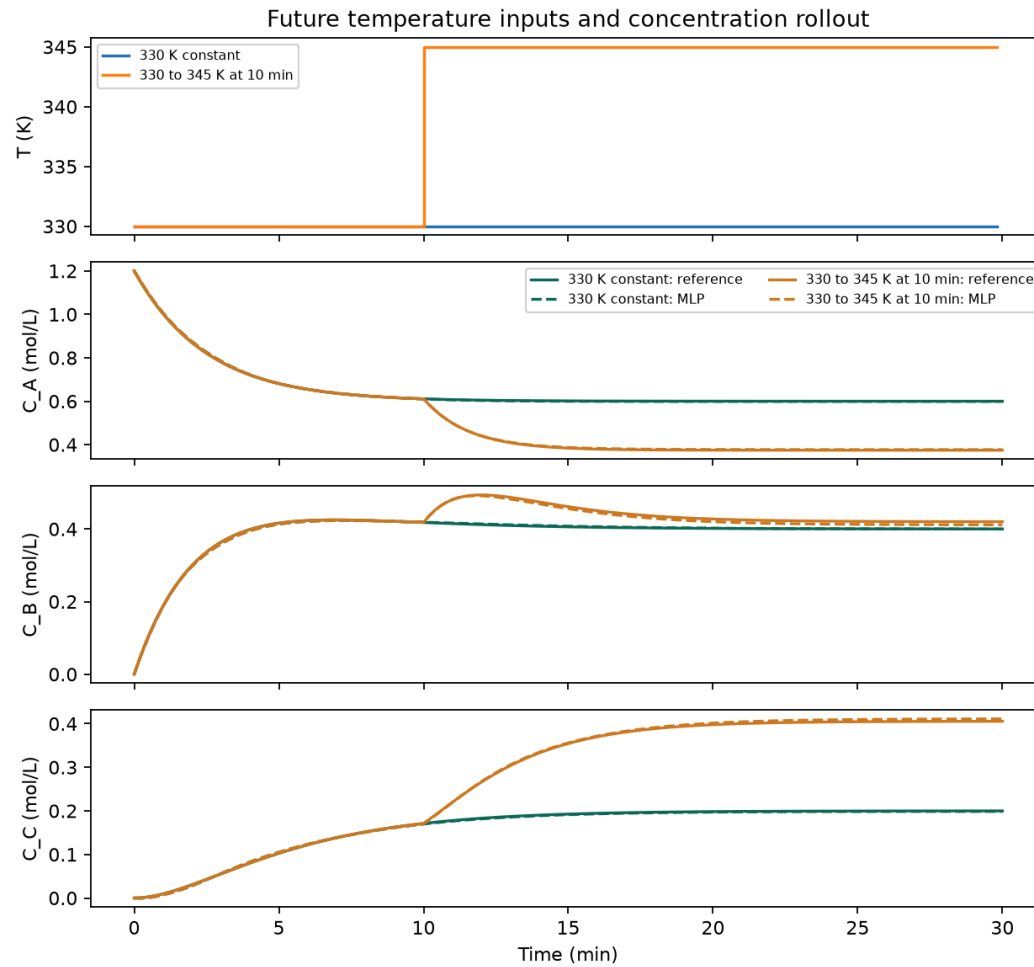
For one-step evaluation, supply the reference concentration at each time. For rollout, begin with the specified initial concentration and then supply the model's previous prediction:

$$\text{one-step: } \hat{c}_{k+1} = f_{\theta}(c_k, u_k)$$

$$\text{rollout: } \hat{c}_{k+1} = f_{\theta}(\hat{c}_k, u_k)$$

The future input sequence remains prescribed in both evaluations. Compare a constant 330 K plan with a plan that changes from 330 to 345 K at 10 min. Both use $\tau = 5$ min, feed = 1.2 mol/L, and initial concentrations [1.2, 0, 0] mol/L.

Two temperature plans: full time series



Solid curves are reference concentrations; dashed curves are MLP rollout. The temperature plan changes all three concentration histories.

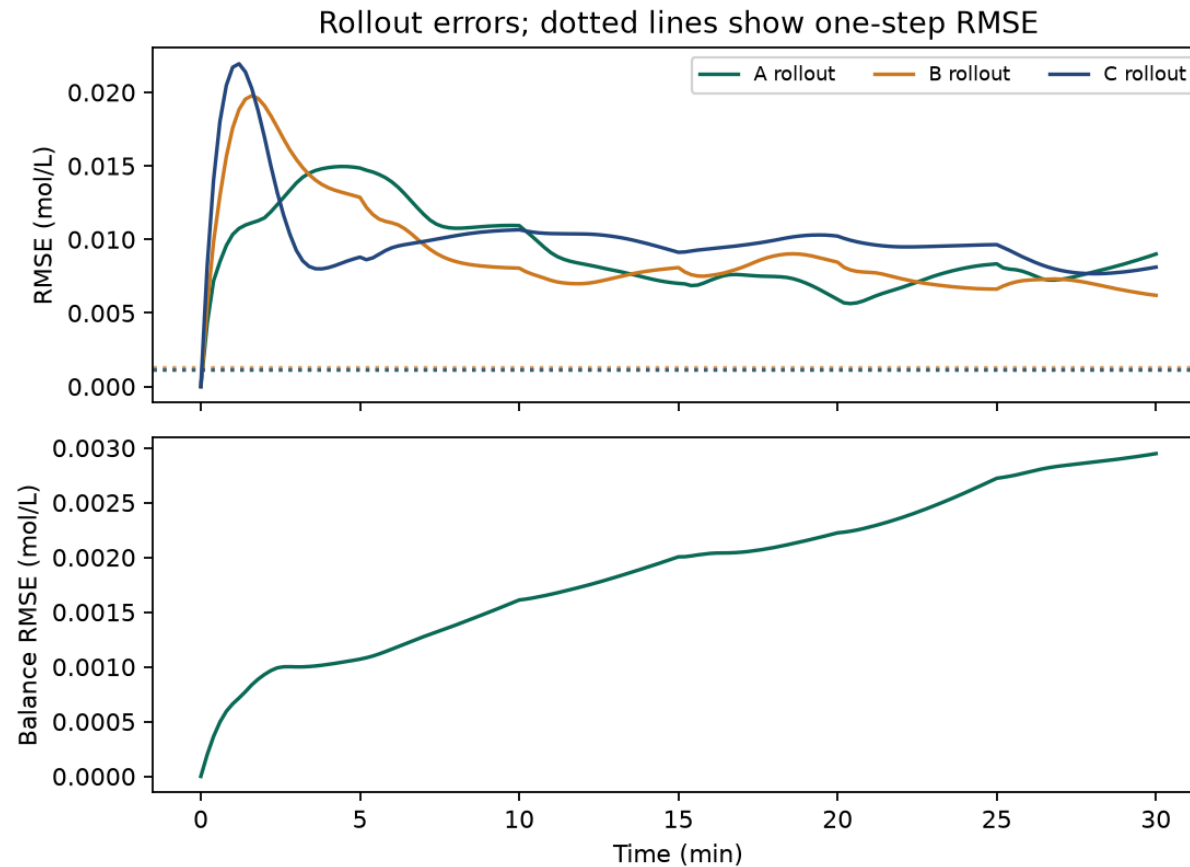
One-step and rollout errors

Test RMSE, mol/L	A	B	C
One-step	0.001170	0.001288	0.001163
30 min rollout	0.009465	0.009629	0.010425

Physical check	One-step	Rollout
Concentration-sum RMSE, mol/L	0.000051	0.002014
Minimum predicted concentration, mol/L	-0.001152	-0.005166

The rollout errors are larger because the next prediction uses an already predicted state. The minimum concentration also reveals small negative predictions. Follow the component errors and concentration sum along the full trajectory.

Follow errors over the prediction horizon



Errors over 20 test trajectories. Dotted lines show the corresponding one-step RMSE.

Compare the final B concentration

Final B concentration at 30 min, mol/L	Reference	MLP rollout
Constant 330 K	0.400091	0.399819
Step from 330 to 345 K	0.420021	0.411526

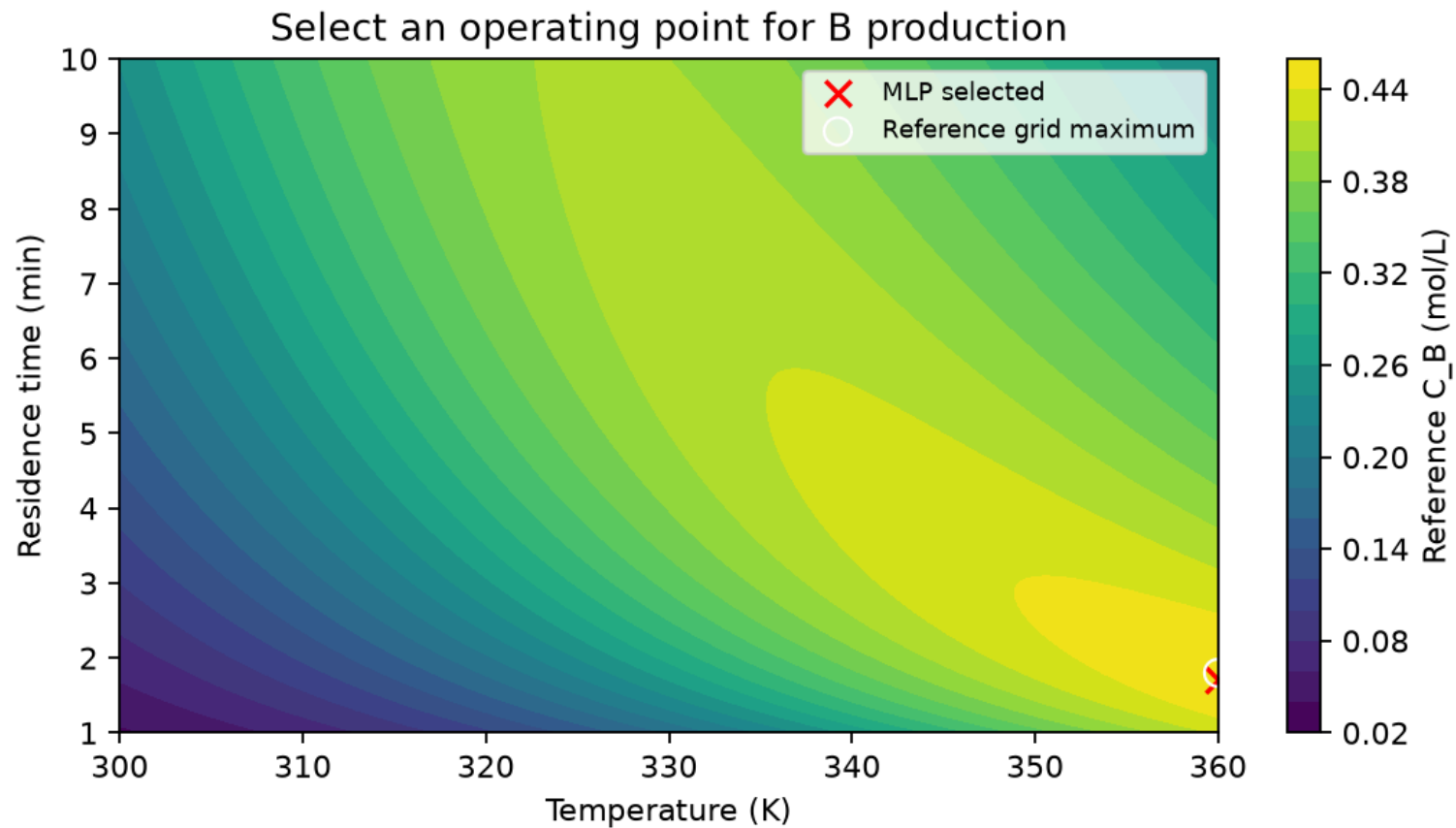
The temperature-step plan gives a higher final B concentration in both calculations. The MLP underestimates the step plan's final concentration; the time-series comparison shows where the discrepancy develops.

Later preview: Select an operating point for B production

Fix the feed at 1.2 mol/L and maximize predicted B concentration. Enumerate temperature in 0.5 K steps and residence time in 0.1 min steps, retaining candidates whose predicted component concentrations lie between zero and the feed concentration. Evaluate the selected point with the reference map.

Quantity	Result
MLP-selected temperature; residence time	360.000000 K; 1.700000 min
Predicted B concentration	0.451111 mol/L
Reference B concentration at this point	0.455285 mol/L
Reference-grid maximizing temperature; residence time	360.000000 K; 1.800000 min
Reference-grid maximum B concentration	0.455382 mol/L
Reference B yield at the MLP-selected point	0.379404

Later preview: Compare the selected operating points



The selected residence time is 1.7 min; the reference grid selects 1.8 min. Compare their reference B concentrations.